

CS4048 Data Science

Project Deliverable 3

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Eye Contact and Engagement Analysis Report

1. Introduction: Project overview and objectives

The project aims at a classification problem. The Columbia Gaze Data Set is used to train models to classify a person in an image as 'attentive', 'distracted' or 'disengaged'.

Dataset: Description and source

- (Columbia Gaze Data Set) Available at : https://www.cs.columbia.edu/CAVE/databases/columbia_gaze/
- Data set contains a total of 5,880 high-resolution images of 56 different people (32 male, 24 female).
- Each image has a resolution of 5,184 x 3,456 pixels.
- 21 of the subjects were Asian, 19 were White, 8 were South Asian, 7 were Black, and 4 were Hispanic or Latino.
- The subjects ranged from 18 to 36 years of age, and 21 of them wore prescription glasses.
- For each subject, images were acquired for each combination of five horizontal head poses (0° , $\pm 15^\circ$, $\pm 30^\circ$), seven horizontal gaze directions (0° , $\pm 5^\circ$, $\pm 10^\circ$, $\pm 15^\circ$), and three vertical gaze directions (0° , $\pm 10^\circ$).
- Total of $5 * 7 * 3 = 105$ combinations for each subject.

2. Methodology: Complete pipeline and all three models

Preprocessing Techniques: Methods used and their intuition

- Resized the images to 224 by 224 since most CNN models requires images to be of this size.
- Pixel values for images were normalized to [0, 1] since most CNN models requires pixel values to be in this range.
- The head pose, horizontal gaze and vertical gaze angles were extracted from image file names and images were labelled as 'attentive', 'distracted', or 'disengaged' based on angle thresholds set by us. This was done to label the dataset for supervised learning.
- The dataset was not balanced with 51% 'distracted' labels, 40% 'disengaged' labels and 8% 'attentive' labels. This imbalance was handled and will be explained later in the report.
- The accuracy of models was not satisfactory in initial training and as a result we had to zoom the pictures by a zoom-factor of 2 so that the eyes and face angle of persons in the images were more highlighted. As a result the accuracy of the models improved in later training.

Training:

- The three models trained are **MobileNet**, **VGG16**, and **DenseNet12**.
- Initially EfficientNet and a few other models were trained but they accuracy was capped at 50% and eventually we abandoned them.
- All the three models were trained in the same manner.
- The data was split in 80:20 ratio for training and testing.
- Since there was class imbalance, we used class weighting which assigned the weight of 3.8 to 'attentive' class, 0.83 to 'disengaged' class and 0.64 to 'distracted' class. This helped in improving the precision, recall and f1 score which low before using class weights.

- The base model was loaded and initially made untrainable.
- Three layers were added to the base model, a pooling layer, a dropout layer and a dense output layer with three neurons since there are three classes. The activation function at the output layer was softmax.
- Initially for head training, the model is compiled with larger learning rate since the head needs complete training.
- For fine tuning the base model is made trainable and compiled again using smaller learning rate since we don't want to disturb the weights in the base model, we just want to fine tune them.
- Curves for accuracy and loss were also plotted.
- Some sample images were outputted to show what predictions were made by the trained model.
- Confusion matrix was also made.
- Accuracy, precision, recall and f1 score were also calculated.

3. Results: Comprehensive model comparison with metrics and visualizations

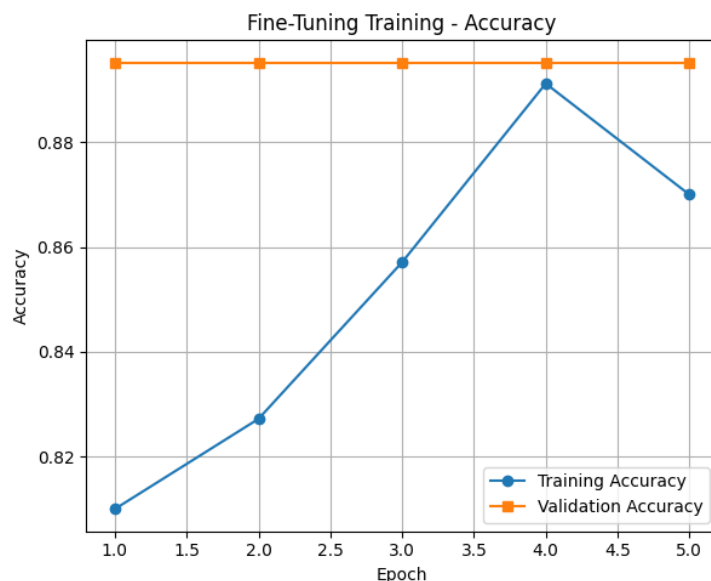
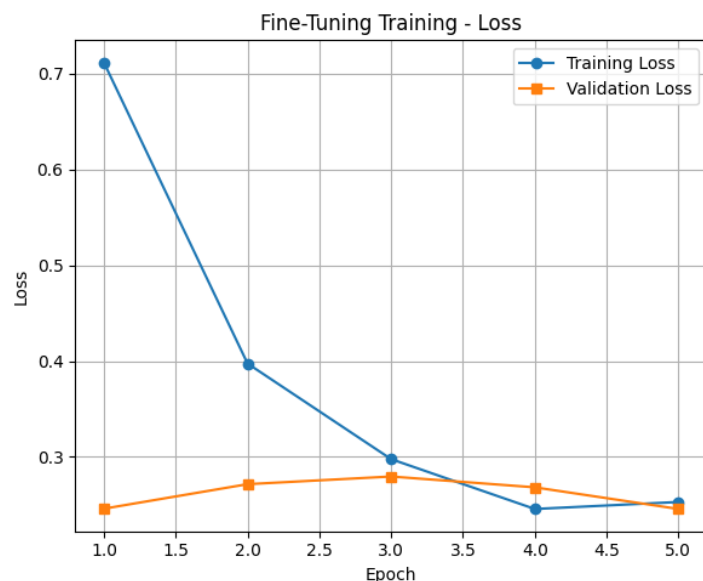
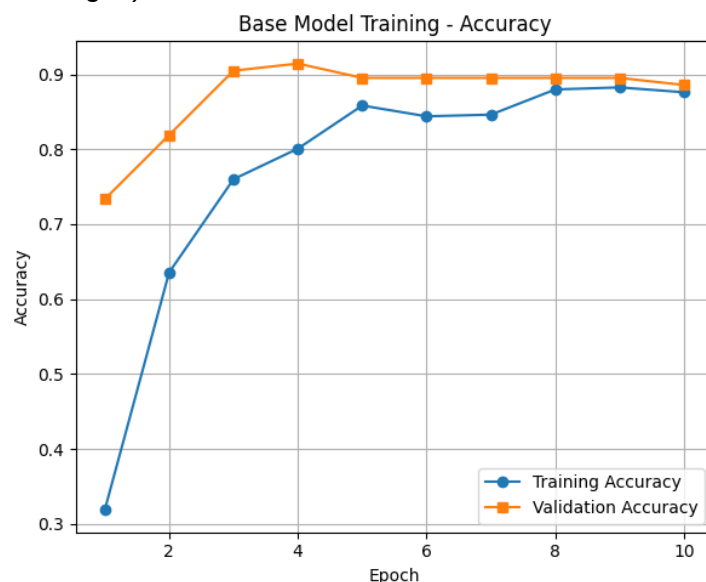
The three models were initially trained on 525 images and the results are:

	MobileNet	VGG16	DenseNet121
Accuracy	89.52%	89.52%	89.52%
Precision	81.67%	79.83%	78.91%
Recall	93.21%	87.04%	83.95%
F1 Score	83.57%	81.70%	80.50%

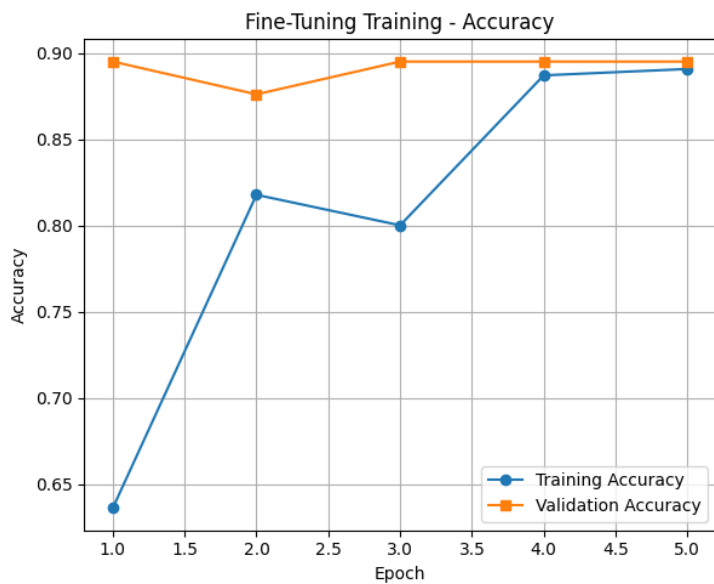
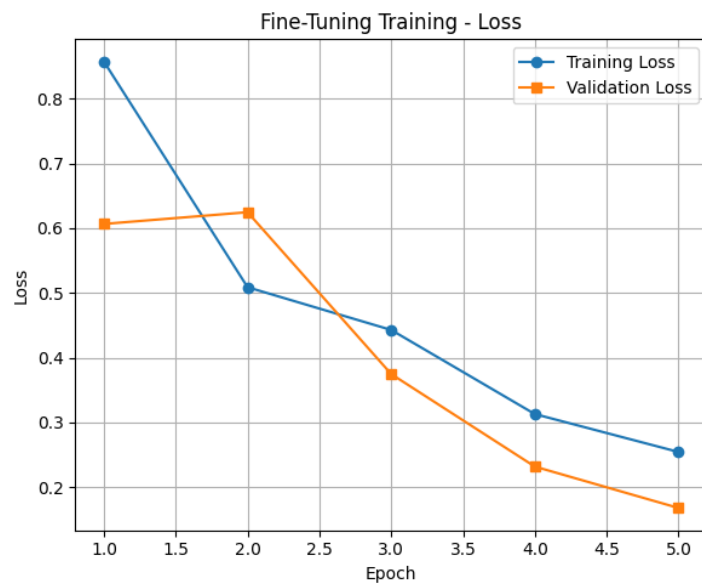
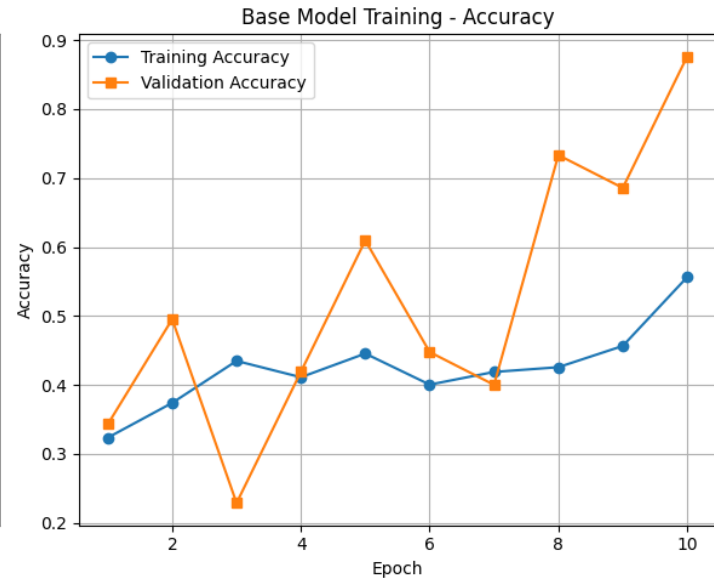
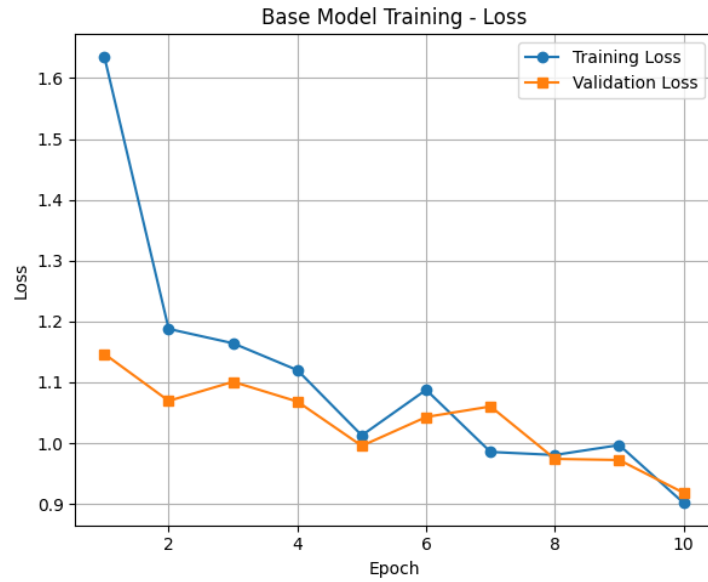
The three models were later trained on 1050 images and the results are:

	MobileNet	VGG16	DenseNet121
Accuracy	88.57%	88.57%	
Precision	80.45%	80.91%	
Recall	91.05%	92.59%	
F1 Score	82.08%	82.64%	

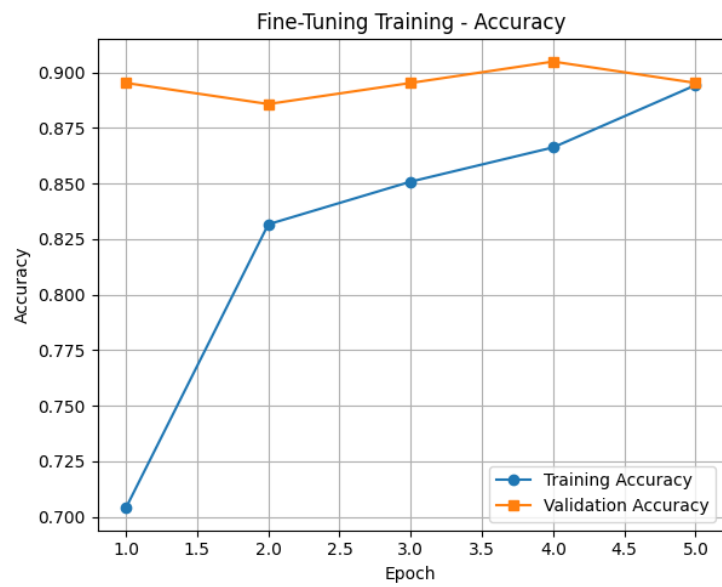
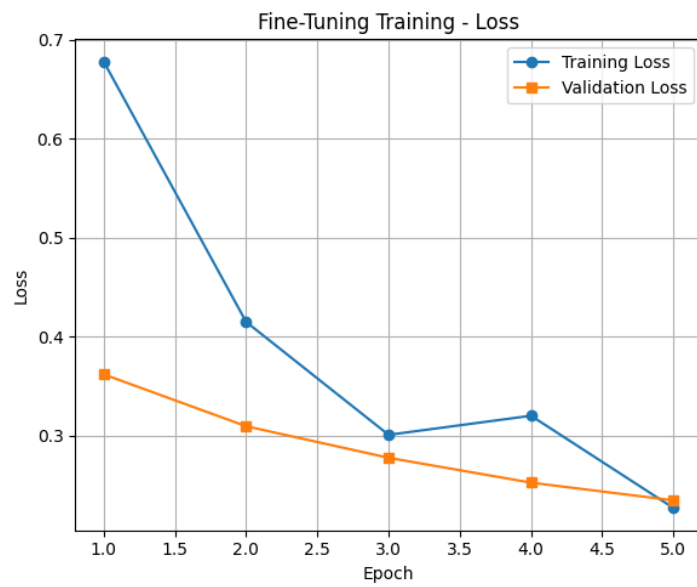
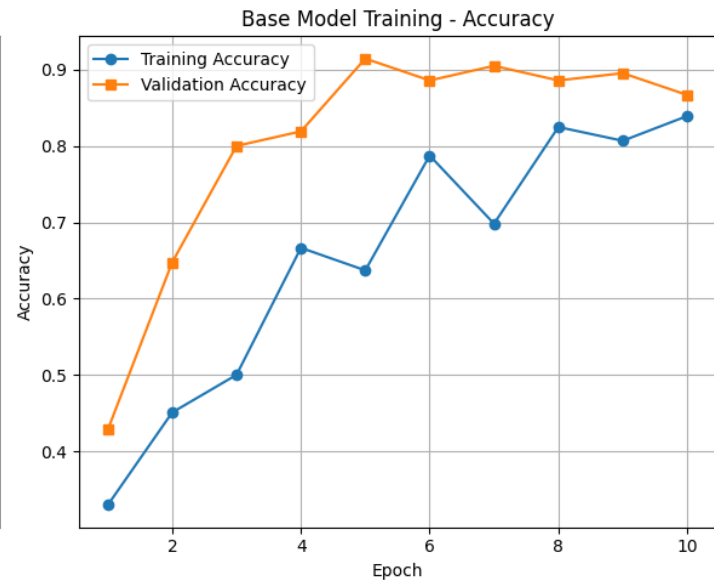
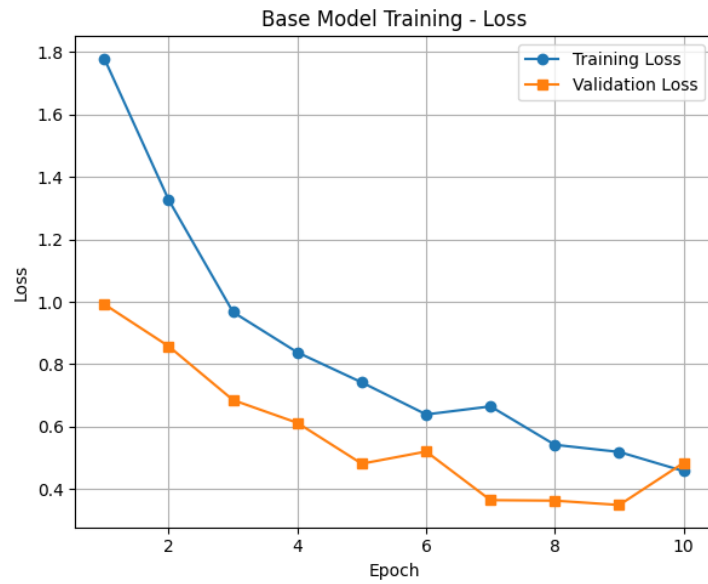
MobileNet (with 525 images)



VGG16 (with 525 images)

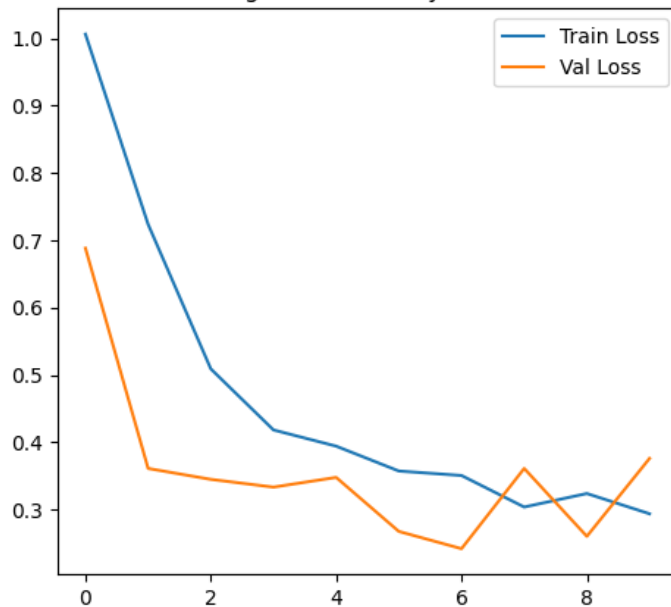


DenseNet121 (with 525 images)

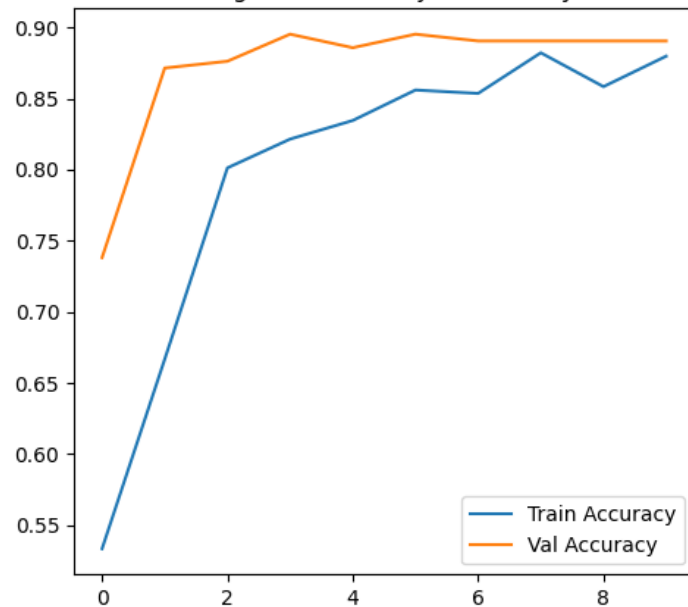


MobileNet (with 1050 images)

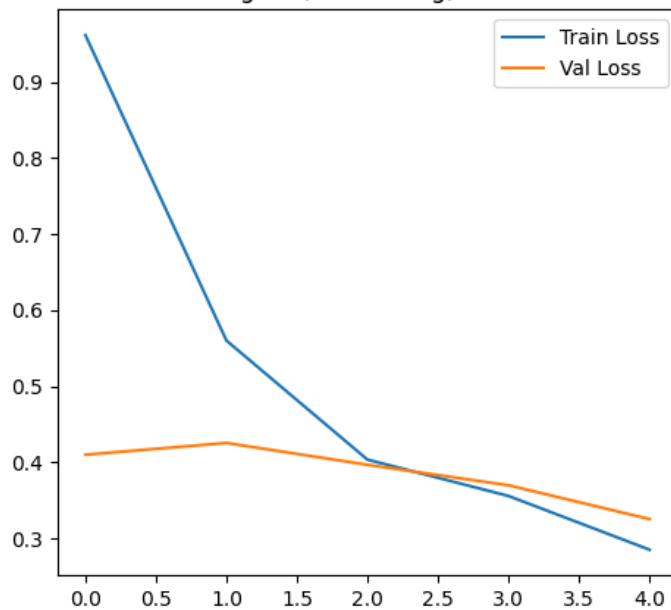
Stage 1 (Head Only) - Loss



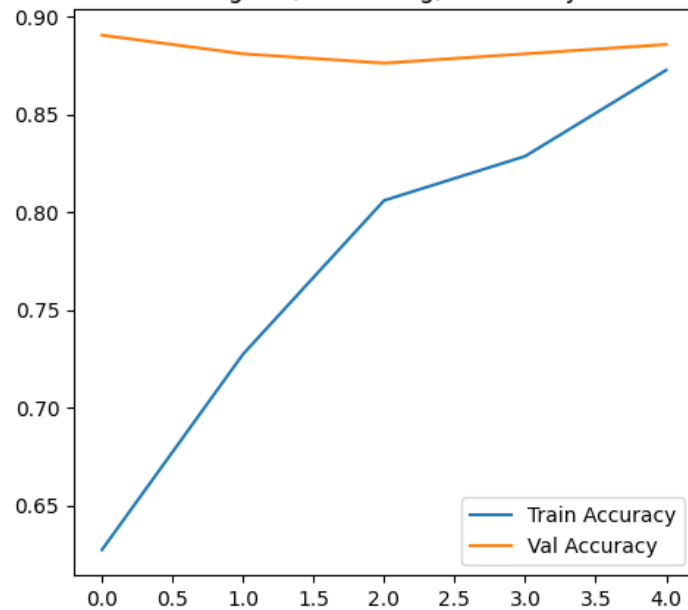
Stage 1 (Head Only) - Accuracy



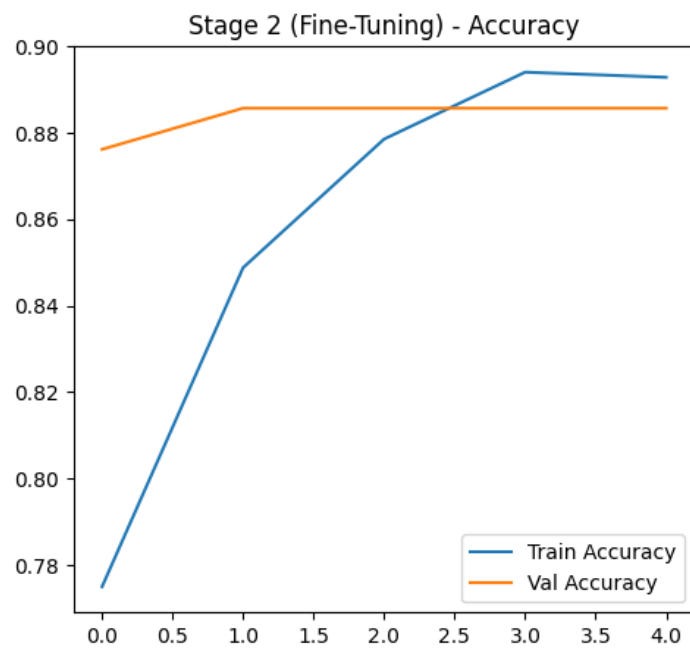
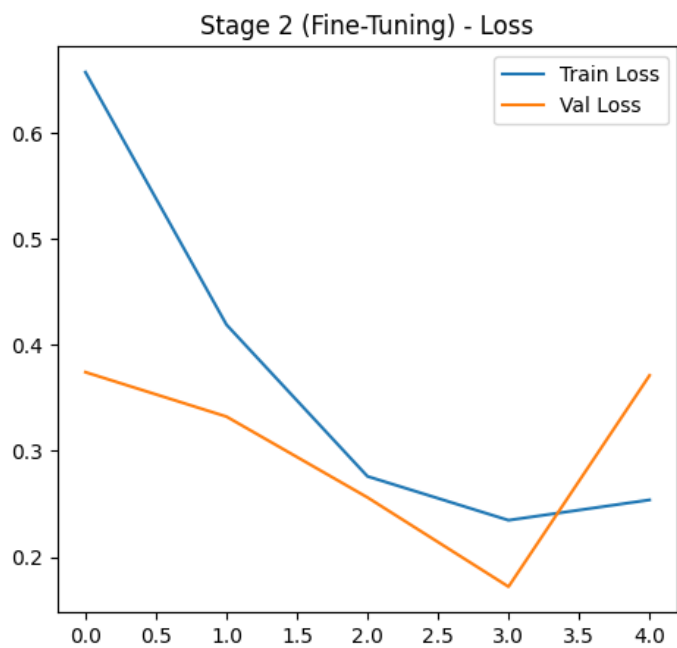
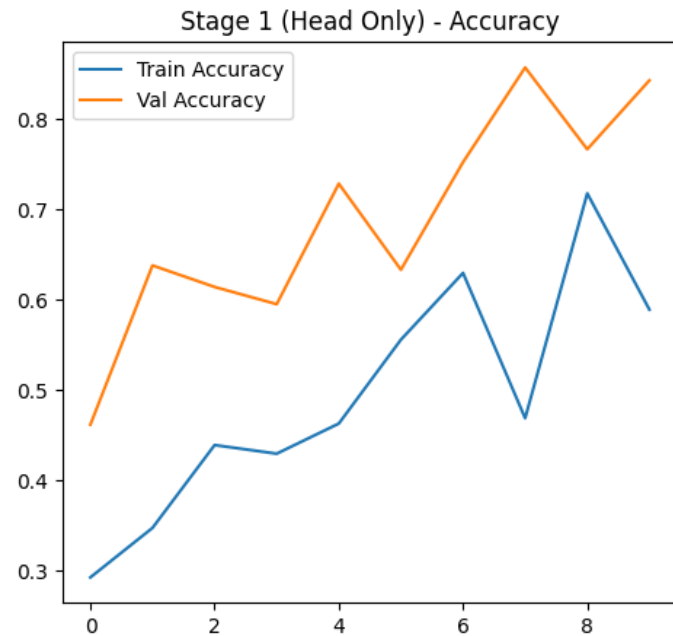
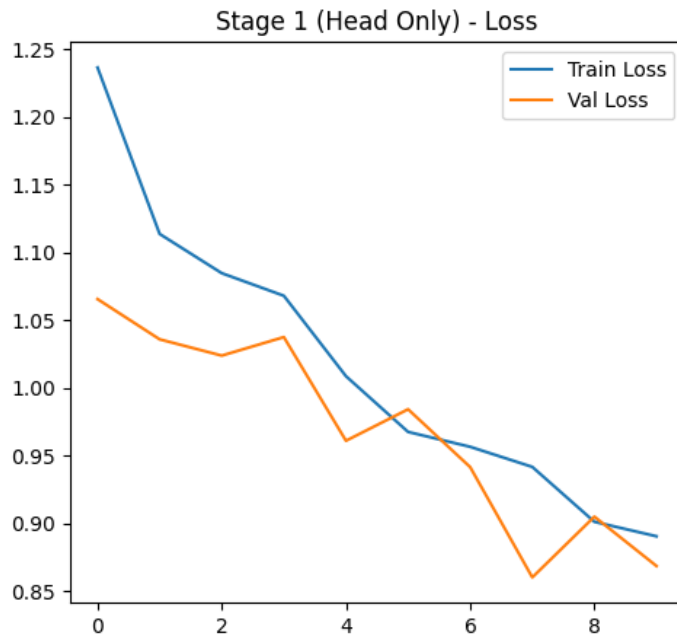
Stage 2 (Fine-Tuning) - Loss



Stage 2 (Fine-Tuning) - Accuracy



VGG16 (with 1050 images)



4. **Discussion:** Analysis, challenges, and solutions
5. **Conclusion:** Key findings, limitations, and future work
6. **References:** All sources used